

Self: Biological Influences

1. [9 marks]

(2013:S3:3b,c)

(a) List **two** functions that are controlled by the

(i) somatic nervous system.

[2]

(ii) autonomic nervous system.

[2]

(b) A person is sitting inside a quiet house watching television when suddenly there is a loud noise outside.

(i) List **three** symptoms produced by the autonomic nervous system that the person might experience. [3]

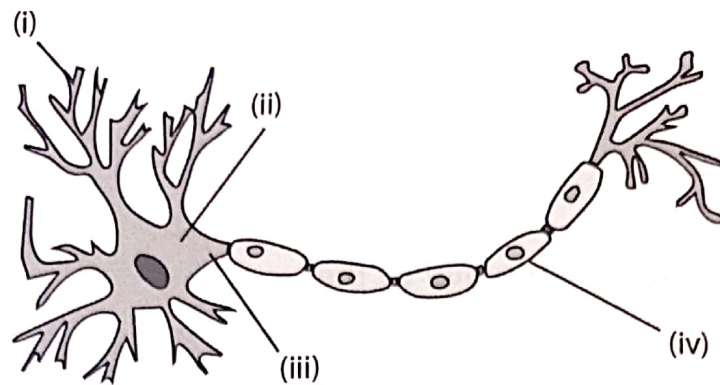
(ii) Name the division of the autonomic nervous system that is responsible for producing these symptoms. [1]

(iii) Name the division of the autonomic nervous system that is responsible for the reduction of these symptoms. [1]

2. [12 marks]

15.53:3a-d)

- (a) Identify the structure of a neuron shown by the labels in the illustration below. [4]



Structure of a neuron

- (i) _____
- (ii) _____
- (iii) _____
- (iv) _____

- (b) Describe the purpose of the structure labelled at (iv) in part (a). [1]

- (c) Describe how neurotransmitters work in communicating messages between neurons at a synapse. [3]

(d) List **two** effects of each of the following neurotransmitters.

(i) Dopamine

[2]

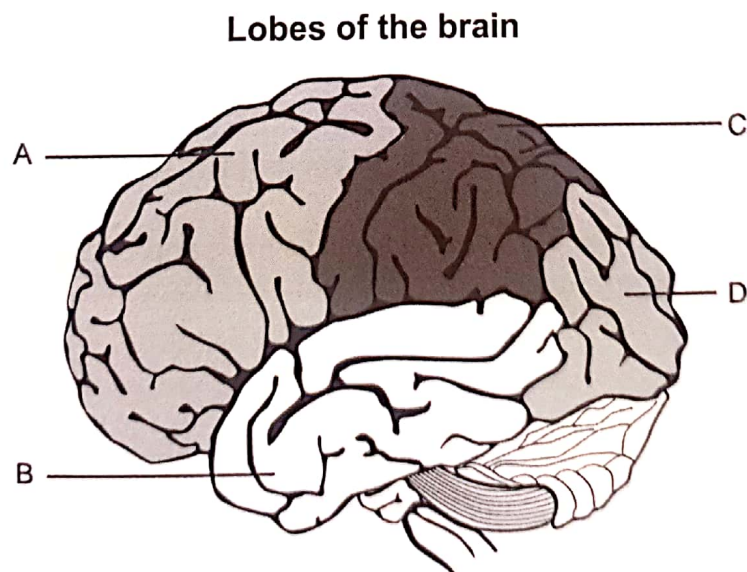
(ii) Serotonin

[2]

3. [18 marks]

(2015:S3:4a-c)

(a) Complete the table below by naming the lobes of the brain shown in the figure below and describing one main function of each lobe. [8]



	Name of lobe	Function
A		
B		
C		
D		

(b) Name the area of the brain associated with each of the following dysfunctions.

- (i) Fluent Aphasia: where a person speaks in long and complex sentences that cannot be understood by others. [1]

- (ii) Inability to understand the body's location in space. [1]

- (iii) Inability to initiate movement. [1]

- (iv) Non-fluent Aphasia: where a person has difficulty in getting words out but is normally understood by others. [1]

- (c) Name **one** physiological and **one** psychological effect of each of the following classes of psychoactive drugs.

(i) Stimulants [2]

Physiological effect: _____

Psychological effect: _____

(ii) Depressants [2]

Physiological effect: _____

Psychological effect: _____

(iii) Hallucinogens [2]

Physiological effect: _____

Psychological effect: _____

4. [15 marks]

(2016:ATAR:4)

- (a) Name the parts of the human nervous system shown by A and B in Figure 1. [2]

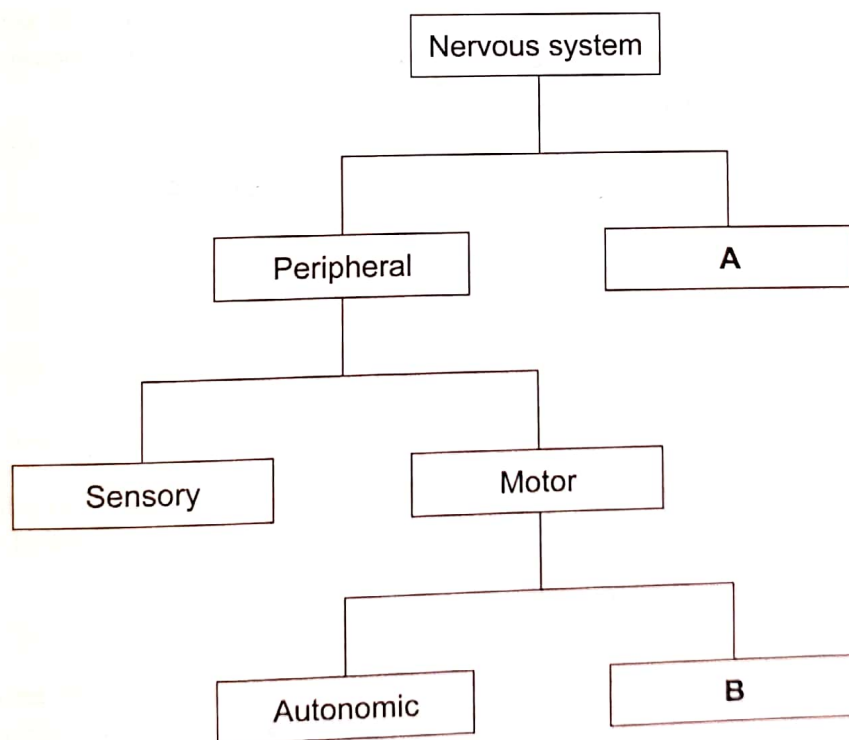


Figure 1. Human nervous system

A: _____

B: _____

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(b) Name **two** examples of physiological responses and identify the part of the autonomic nervous system responsible for these responses in the following situations.

(i) Meditating in a quiet location. [3]

Physiological response one: _____

Physiological response two: _____

Part of autonomic nervous system: _____

(ii) Discovering a large spider crawling in your hair. [3]

Physiological response one: _____

Physiological response two: _____

Part of autonomic nervous system: _____

(c) List **two** functions for which the left hemisphere of the brain is responsible. [2]

One: _____

Two: _____

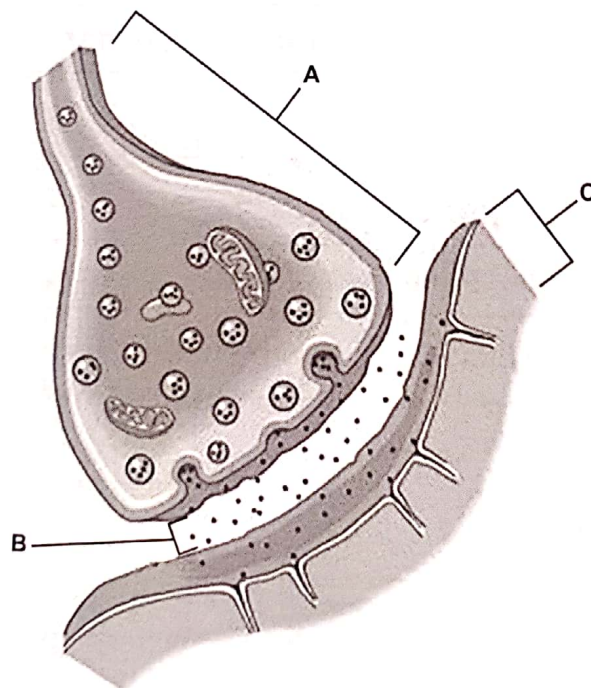


Figure 2. Diagram of a synapse

(d) (i) Name the structures shown at A and C in Figure 2. [2]

A: _____

C: _____

- (ii) State what occurs at A, B and C in the process of neural transmission. [3]

A: _____

B: _____

C: _____

5. [15 marks]

(2017:ATAR:3)

Stacey got up during the night to get a glass of water. While walking to the kitchen, she felt something brush against her leg. She did not realise that it was her cat and was so scared that she tripped and fell, hitting her head on the wall. Stacey's mother woke up and found her daughter lying on the floor. When she asked Stacey what had happened, Stacey had trouble in hearing her and asked her mother to repeat herself. Her mother also noticed that Stacey was not formulating sentences correctly or making any sense.

- (a) (i) Name the lobe of Stacey's brain that has most likely suffered trauma and explain the reason for your response, using evidence from the scenario. [3]

- (ii) If Stacey woke up and was unable to process visual information, name the lobe of the brain that might have been affected. [1]

- (iii) Name the lobe of the brain responsible for processing the feeling of the cat brushing against Stacey's leg. [1]

- (b) When Stacey felt the cat brush against her leg, her body produced a hormone in response to the situation.

- (i) State the name of this hormone.

[1]

- (ii) State **two** physiological effects of this hormone.

[2]

One: _____

Two: _____

- (c) (i) Name the branch of the nervous system that is involved in transmitting information to and from the spinal cord.

[1]

- (ii) Name the **two** types of neurons that are involved in transmitting information to and from the spinal cord.

[2]

One: _____

Two: _____

- (d) Alcohol and marijuana are two examples of psychoactive drugs. Complete the table below by stating the class of each drug and **one** effect each of these drugs might have on a person's thoughts or behaviour.

[4]

Psychoactive drug	Class of drug	Effect on thoughts or behaviours
Alcohol		
Marijuana		

6. [10 marks]

(2018:ATAR:4)

For copyright reasons this image cannot be reproduced in the online version of this document, but may be viewed at http://esmartkids.epcor.ca/websites/sse_ca/body/nervous.html

- (a) What is represented at 'A' in the diagram above? [1]

- (b) (i) Neurotransmitters are released from axon terminals. Describe the function of neurotransmitters in the process of neural transmission. [2]

- (ii) Name **two** neurotransmitters and state **one** effect of each. [4]

Neurotransmitters	Effect of neurotransmitters

- (c) Fiona's mother has a condition that causes her body to fail to produce a particular neurotransmitter correctly. Fiona has now discovered that she has the same condition. Explain how it is possible that they share this condition. [3]

7. [14 marks]

(2019:ATAR:4)

- (a) (i) Name the **two** components of the central nervous system.

[2]

One: _____

Two: _____

- (ii) State the functions of the central nervous system and peripheral nervous system.

[2]

	Function
Central nervous system	
Peripheral nervous system	

Simon is walking home from work late one evening. He hears footsteps coming up close behind him which causes him to feel scared.

- (b) (i) What is the role of the sympathetic nervous system in this situation?

[1]

- (ii) State **three** physiological changes Simon's sympathetic nervous system would initiate in this situation.

[3]

One: _____

Two: _____

Three: _____

As Simon continues walking, the footsteps stop as the person behind him turns down another street. He no longer feels scared.

- (iii) What is the role of the parasympathetic nervous system in this situation?

[1]

Jill is a 25-year-old and decides to go out to a night-club with her friends. After a few hours, Jill's friends notice her acting strangely. She starts babbling about how everyone should get out of the mud now because lizards in golf shoes are coming to eat them. She becomes agitated and panicky as she tells her friends about what is happening. They suspect she has taken some kind of drug.

- (c) (i) Define the term 'psychoactive drug'. [2]

- (ii) What class of psychoactive drug is likely to be causing Jill's strange behaviour? [1]

- (iii) Justify your answer for part (c)(ii) using evidence from the scenario. [2]
